

Synergistic Activities — Aaron Cherian

1. **Open-source research tooling for data quality and benchmarking.** Designs and builds diagnostics and benchmarking tools for the FreeMoCap platform, giving the research community practical instruments for assessing the quality and reliability of markerless motion-capture data across systems and software versions.
2. **Validation research and publication.** Conducts validation studies of open-source markerless motion capture against established marker-based standards and publishes the results in the research literature, providing the community with openly available evidence about the capabilities and limitations of accessible motion-capture methods.
3. **Collaborative research across application domains.** Co-authors research papers with external researchers applying motion capture in their own domains, supporting the adoption of accessible movement measurement in fields far from the original development team's expertise and transferring methods knowledge to new research communities.
4. **Interface regularization and contributor onboarding.** Works to regularize and document the project's application programming interfaces and validation strategies, explicitly to lower the technical barriers for outside developers and researchers to contribute to the ecosystem — transferring architectural knowledge beyond the core development team.
5. **Community teaching and mentoring.** Teaches and mentors students, early-career researchers, and professionals through the FreeMoCap community channels, providing guidance on motion-capture methodology, data quality, and open research practice.